

FPV CARD BOSS

The festival FPV stabilizing, delivery & slate command center - on Windows **and** Mac, with a live mobile companion for the field.

Operator Manual - Setup Guide - SOP

Version 1.2.0 - Windows & macOS

TWO WAYS TO READ THIS

New to the job? Read straight through - Section 1 explains the work before the buttons. **Done it before?** Jump to the Quick Reference at the back.

1 The Job & The SOP

What you're actually doing out here, and why.

You are the **stabilizer operator** for an FPV drone team at a music festival (EDC Las Vegas, Beyond Wonderland, and the like). Pilots fly all night and hand you SD cards - often at 3 AM, out of order, several at once. Your job: get every card's footage **copied, stabilized, and delivered** to the right places before the editors need it, and logged so nothing is lost.

The one rule that never breaks

Never destroy the originals. Footage always lives in two halves: **RAW** (untouched clips straight off the card) and **STABILIZED** (the smoothed exports), in separate subfolders per shot - so originals are never overwritten and you can always re-export from RAW.

Why stabilize, and why a robot

FPV drones are tiny and twitchy; raw footage is unusably shaky. GoPro Player's HyperSmooth Pro (ReelSteady) flattens that motion. Every clip from every pilot must use the SAME settings so footage cuts together cleanly - that consistency is the whole reason this app exists. The app physically drives GoPro Player (a robot) so the settings are identical every time.

WHERE FOOTAGE GOES

Local - your fast working drive. **Media Drive** - the master archive (RAW + STABILIZED together, named by card ID). **Bella / Social** - stabilized clips only, named by artist (social never gets raws). Then it's logged in the shared **Media Master** sheet.

THE WHOLE LOOP IN ONE BREATH

Card in hand - make folders - copy card to RAW - robot stabilizes - move exports to STABILIZED - copy to Media + Bella - log it - wipe the card - next.

2 Requirements, Install & Platform

What you need, and telling the app which computer it's on.

Requirement	Detail
Computer	A Windows 10/11 PC or a Mac (Apple Silicon or Intel, macOS 10.14+), with a GPU that decodes HEVC.
GoPro Player	GoPro Player + HyperSmooth Pro (ReelSteady) installed, with the HyperSmooth Pro unlock.
Drives	A fast local drive, plus the Media and Bella/Social drives or network volumes.
Phone	iPhone or Android for the mobile companion (optional). Tailscale app for field use.

Choosing your platform

On first launch the app asks **Which computer is this - Windows PC or Mac?** and pre-selects what it detects. Your choice decides which engine drives GoPro Player and your drives. Change it any time from the platform badge in the corner.

MAC: TWO ONE-TIME PERMISSIONS

The first time the robot runs on a Mac, macOS asks for **Accessibility** (to move the mouse/keyboard) and **Screen Recording** (so calibration can read the screen). Approve both in System Settings > Privacy & Security.

WHAT DIFFERS BETWEEN WINDOWS & MAC

Nothing you can see: same workflow, buttons, settings, slate, and companion. Under the hood, copying uses Robocopy on Windows and rsync on Mac, the robot uses a different click engine, and GoPro exports land in Videos (Windows) vs ~/Movies (Mac). Calibration is saved separately per computer.

3 First-Time Setup

Do this once on each computer. Open Setup.

- 1. Set the folder paths.** Local root, Media drive, Bella drive, Raw Dump folder, SD Card drive/volume, and the GoPro output folder.
- 2. Add your pilots.** Each gets a card prefix + starting number (e.g. L from 1 produces L_001, L_002). The active pilot sets the next card ID and filters the shot list.
- 3. Calibrate the GoPro robot.** Run Calibrate GoPro Robot and follow the 14 steps - hover each control and press SPACE (Section 5).
- 4. Set the phone password** (optional). Gates only the Move Files section on the phone; Shot List & Slate stay open.

CALIBRATION IS PER-COMPUTER & PER-RESOLUTION

Saved for this machine at its exact resolution. Re-calibrate on a new computer, if you change scaling/resolution, or if you move/resize the GoPro window. Windows and Mac calibrations are independent.

Folder structure the app builds

Local: [Local]\[Event]\[Pilot]\Day[N]\[Artist]\RAW + \STABILIZED
Media: [Media]\[CardID]\ (RAW + STABILIZED together)
Bella: [Bella]\[Artist]\ (STABILIZED clips only)
Dual: ... \STABILIZED\HORIZON LOCK\ (2nd pass, Dual Mode)

4 The Core Workflow

Festival mode - from card in hand to delivered & logged.

1. **Pick the card's assignment.** Confirm pilot + day, pick the artist/shot (Choose From List when cards arrive out of order). Sets the card ID and folder names.
2. **Create Directory Paths.** Makes RAW & STABILIZED locally, on the Media Drive, and the artist folder on Bella.
3. **Copy SD Card to Local RAW.** Copies the card in and verifies count + size. If RAW already has files, it uses a fresh BATCH_02 subfolder so the robot only stabilizes new clips.
4. **Set export options.** Toggle Horizon Lock or Dual Mode if needed (Section 6).
5. **Auto-Run GoPro Batch.** The robot drives GoPro Player. **Hands off the mouse and keyboard until it finishes.**
6. **Move Files to STABILIZED.** When the export finishes, move the new clips in (only files created after the robot started).
7. **Deliver.** Copy to Media Drive, Copy to Bella, Dump Raws as needed.
8. **Complete Card.** Logs it, advances the number, clears for the next. Then update the Media Master sheet.

FROM THE COUCH

Every step from Move Files on can be triggered from your phone, or run hands-free with Auto mode (Section 6).

5 Stabilization & Calibration

The locked-in look, and teaching the robot your screen.

The SOP export settings - every clip, every pilot

Setting	Value
Codec	HEVC (H.265) 10-bit
HyperSmooth Pro	ON
Smoothness	15
Cropping	15
Aspect Ratio	8:7 (widest field of view)

WHY UN-GAIN MATTERS

Smoothness and Cropping are linked by default - moving one drags the other. The robot first clicks the un-gain / chain-link button to unlink them, so it can set each to 15 independently. That's why it's a calibration point.

Calibration - 14 points (hover each, press SPACE)

- > batch list - 10-bit - HyperSmooth - un-gain - Horizon Lock
- > Smoothness start & end - Cropping start & end
- > aspect-ratio open - 8:7 option - drop zone - Start - Remove

RE-CALIBRATE WHEN THE WINDOW MOVES

The robot clicks fixed screen positions. Move/resize the GoPro window, change resolution, or switch computers - re-calibrate or it misses.

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Auto - Horizon Lock - Dual Mode

Hands-free delivery, a level horizon, and both versions at once.

Auto vs Manual

- > **Manual** (default) - you click Move, each delivery, and Complete yourself.
- > **Auto** - on export finish it auto-moves to Media to Bella, completes the card and advances. Stops and alerts on any failure; never completes a card that didn't fully deliver.

RAWS ARE ALWAYS MANUAL

Dumping raws is never part of Auto mode or the phone's Send to All - a deliberate click, unless you enable Auto-Dump. Always dump raws before wiping a card.

Horizon Lock

Toggle ON (turns blue) before running a batch and the robot enables Horizon Lock so footage exports level. If on but not calibrated, the app warns and skips it - re-calibrate to enable.

Dual Mode - both versions in one run

The robot exports every clip **twice**: a normal pass into STABILIZED, then a second pass with Horizon Lock into a STABILIZED\HORIZON LOCK subfolder. Editors get the regular and the level-horizon version with no filename collisions. About twice as long; needs the Horizon Lock calibration point.

WHEN TO USE DUAL

Use it when the team hasn't decided between locked or unlocked horizon for a set - deliver both and let them choose in the edit.

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The Mobile Companion

Run the shot list, slate, and deliveries from your phone.

Connecting - three addresses

In Setup, under Open on your phone, scan a QR code or tap Copy, then Add to Home Screen.

- > * **Tailscale (HTTPS)** - <https://...ts.net>. The best one: works anywhere with Tailscale on, opens Shot List & Slate even with the computer OFF, and enables the slate mic.
- > **Tailscale (HTTP)** - works anywhere with Tailscale on, but online-only.
- > **Same Wi-Fi (LAN, 192.168...)** - quick access on the same Wi-Fi. Online-only.

WHY HTTPS IS SPECIAL

Browsers only allow offline caching and microphone access over a secure (HTTPS) address - the Tailscale HTTPS link. Open it once while the computer is running (Tailscale on) so the phone caches a copy; after that it works offline.

The sections & delivery

- > **Shot List & Slate** - open to anyone; works offline.
- > **Simple Slate** - the slate on its own.
- > **Site Map** - the venue map image (works offline once cached).
- > **Move Files** - live status & delivery; needs the password.

DELIVERY FROM THE PHONE

Auto - Send to All is one tap that moves files and copies to every set-up drive (Media + Bella). After raws are backed up, a Delete SD Card button appears so you can finish a card entirely from your phone.

8 Shot List, Slate & Site Map

Plan the night, slate the shots, find your way around.

The shot list

Fill it three ways: **Load Sample EDC** (practice), **Import CSV** (the production's real list - drag a .csv onto the window; columns Artist, Stage, Festival, Pilot, Day), or **add by hand**. Each shot is pending, completed, or skipped. Marking Done/Skip on the phone updates the same row on the computer and fires a red blink + ding there. Skipped shots drop out of the next-card queue.

LIVE AUTO-SYNC - NO IMPORT STEP

The phone keeps the shot list in sync with the computer automatically. Change the list on the PC and connected phones update within a few seconds. Anything already marked completed or skipped stays that way - only new or future shots change.

The festival slate

- > **GoPro Labs sync QR** - scan with a GoPro to sync clock/timecode (Stable, Med, Fast, Freeze; live TC for 24/25/30/60 fps).
- > **Take counter** - bumps save back to the linked shot.
- > **Themes, landscape lock**, and a clean **fullscreen** slate for the camera.

MIC & SCREEN-WAKE NEED HTTPS

Audio take-recording and keeping the screen awake only work over the Tailscale HTTPS address. Plain http:// blocks them.

Site map

In Setup > Mobile Dashboard, add a venue map image (PNG/JPG/WEBP/GIF/BMP/SVG). It shows full-screen on the phone, works offline once cached, and updates on every phone instantly when you replace it.

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Simple Mode & Troubleshooting

The lighter flow, and quick fixes.

Simple Mode

Switch to Simple mode for one-off or non-assignment work. Instead of picking an artist from a shot list, you type a folder name; the app makes RAW/STABILIZED under Local\Show\[Folder], runs the same SD-copy and robot, and copies to Media/Bella by toggle. No shot-list queue, no Dump Raws, no Complete-and-advance.

Troubleshooting

Symptom	Fix
Robot clicks the wrong spots	Re-run Calibrate GoPro Robot (tied to this computer's resolution/scaling and the GoPro window position).
Mac robot does nothing / errors	Grant Accessibility + Screen Recording in System Settings > Privacy & Security, then re-run.
File count doesn't match	A clip may have been missed - re-run rather than delivering an incomplete card.
Phone says Offline	App must be open on the computer; same Wi-Fi or Tailscale on. Shot List & Slate still work offline.
Phone marks don't reach the PC	Make sure the phone is connected (Tailscale / same Wi-Fi). The list auto-syncs every few seconds; marks queue and retry. No import step.
New computer	Copy the app, redo Setup - paths, calibration & password are per-computer.



Quick Reference

The whole job on one page, for when you've done it before.

THE LOOP

Pick assignment - **Create Folders** - **Copy SD to RAW** - set Horizon/Dual - **Auto-Run GoPro** (hands off) - **Move Files** - **Copy Media** + **Copy Bella** - **Dump Raws** - **Complete Card** - update Media Master - wipe card.

Export settings

HEVC 10-bit - HyperSmooth Pro ON - Smoothness 15 - Cropping 15 - Aspect 8:7. Un-gain unlinks the sliders first.

Calibration (14)

batch - 10-bit - HyperSmooth - un-gain - Horizon Lock - Smooth start/end - Crop start/end - aspect open - 8:7 - drop zone - Start - Remove.

Folder map

Local: Event\Pilot\Day\Artist\RAW+STABILIZED - Media: [CardID]\ - Bella: [Artist] (STAB only) - Dual: STABILIZED\HORIZON LOCK\

Modes

Horizon Lock = one level export - Dual = regular + Horizon (2x time) - Auto = move to Media to Bella to complete (raws manual).

Phone connect

Tailscale HTTPS = offline + mic (Add to Home Screen) - Tailscale HTTP / LAN = online only - open once online to cache.

Media Master sheet

Tab FILM CREW - B = Card ID - C = Size GB - I = Artist - J = COMPLETED. Update only after Media has the full folder.

MAC ONE-TIME

Grant Accessibility + Screen Recording. Calibration is separate from Windows.

NEVER DELIVER SHORT

If the file count doesn't match, re-run. Re-calibrate whenever the GoPro window moves.